

Supplementary Information for Generalized Ideal Point Models

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1 Additional Likelihood Details

In the main text I have followed convention in assuming that the observed data Y_{ijt} is a binary outcome (Bernoulli distribution). However, this is a limitation which unnecessarily restricts dynamic ideal point models to binary data. There are a number of implementations of mixed

outcomes for static ideal point models, such as the Poisson “Wordfish” model (Slapin and Proksch 2008), the ordinal model in Imai et al. (2016), and the multinomial model in Goplerud (2019). The restriction of dynamic models to binary outcomes requires either dichotomization of the observed indicators or for scholars to employ a static model that obscures over-time variation.

Within the Bayesian framework I have proposed, and combined with the extraordinary flexibility of Stan’s HMC engine, virtually any distribution for Y_{ijt} is possible and it is even possible to let the distribution for Y_{ijt} to vary at the item level, i.e., to include items of binary, ordinal, or even continuous distributions for the same latent variable. These types of mixed outcomes are especially important for survey data in which questions could have binary, ordinal, proportions, counts or continuous response distributions.

To use a different distribution, the likelihood function $L(\cdot)$ in main text Equation 2 can be modified with any probability density or mass function that properly integrates or sums to 1 where m is an index for all M distributions used for the mixed outcome Y_{ijtm} :

$$L(Y_{ijtm}|\alpha_{it}, \gamma_j, \beta_j) = \prod_{i=1}^I \prod_{j=1}^J \prod_{t=1}^T \prod_{m=1}^M L_m(\gamma_j \alpha_{it} - \beta_j) \quad (1)$$

where L_m represents the likelihood function for the m th distribution. All of these models can also be estimated marginal of missing data r . Below I provide more details about the different possible likelihoods in `idealstan`.

1.1 Ordinal Derivations: Rating Scale and Graded Response

There are two possible ordinal models that can be implemented in the ideal point framework. The first is known as a rating-scale model in the IRT framework and is structurally similar to the ordered logit model. Imai et al. (2016) were the first to implement this model for ideal points using an EM algorithm.

Following the notation I introduced, a rating-scale model for $k \in K$ outcomes can be modeled as follows for observed data:

$$L(Y_{ijtk}|\alpha_{it}, \gamma_j, \beta_j) = \prod_{i=1}^I \prod_{j=1}^J \prod_{t=1}^T \begin{cases} 1 - \zeta(\gamma_j \alpha_i - \beta_j - c_1) & \text{if } K = 0 \\ \zeta(\gamma_j \alpha_i - \beta_j - c_{k-1}) - \zeta(\gamma_j \alpha_i - \beta_j - c_k) & \text{if } 0 < k < K, \text{ and} \\ \zeta(\gamma_j \alpha_i - \beta_j - c_{k-1}) - 0 & \text{if } k = K \end{cases}$$

where again $\zeta(\cdot)$ represents the logit function. In this version, each ordinal category k less one is assigned a cutpoint c_k . In ideal point terms, these cutpoints divide the ideal point space across categories. A direct application is to Likert scales such as those commonly found in

survey data. Importantly, because `idealstan` allows for mixed outcomes at the observation level, the number of ordinal categories K can likewise vary across items.

A different parameterization is known as the graded response model in IRT. This model allows the cutpoints c_k to vary by item. As a result, instead of including separate cutpoints, we can change the model by indexing each item-specific intercept for each category k :

$$L(Y_{ijtk} | \alpha_{it}, \gamma_j, \beta_j) = \prod_{i=1}^I \prod_{j=1}^J \prod_{t=1}^T \begin{cases} 1 - \zeta(\gamma_j \alpha_i - \beta_{jk-1}) & \text{if } K = 0 \\ \zeta(\gamma_j \alpha_i - \beta_{jk-1}) - \zeta(\gamma_j \alpha_i - \beta_{jk}) & \text{if } 0 < k < K, \text{ and} \\ \zeta(\gamma_j \alpha_i - \beta_{jk-1}) - 0 & \text{if } k = K \end{cases}$$

Though not employed in political science to any great extent, this alternative formulation to ordered logit permits more flexible inference on the cutpoints. Allowing cutpoints to vary by item will enable much more accurate prediction of categories with fewer numbers of observations. It also increases information about the items/bills, though of course at a cost of including K additional parameters per item. My intention in including it here is not to suggest that it is a superior model to the standard rating-scale approach, but rather as an alternative that may be useful in some situations.

1.2 Poisson Distribution

The Poisson distribution has a considerable history as a latent variable model for word counts (Slapin and Proksch 2008), though it can be more broadly applied to any unbounded positive discrete variable. For `idealstan`, the Poisson parameterization is as follows:

$$P(Y_{ijtk} | \lambda_{ijtk}) = \frac{\lambda_{ijtk}^{Y_{ijtk}} e^{-\lambda_{ijtk}}}{Y_{ijtk}!} \quad (2)$$

$$\lambda_{ijtk} = e^{\gamma_j \alpha_i - \beta_j} \quad (3)$$

where λ_{ijtk} is the mean count and contains the IRT linear model with the exponential link function.

1.3 Normal and Log-Normal Distributions

The next two distributions have not been widely used in the ideal point literature but have been commonly deployed in continuous latent variable methods like principal components and factor

analysis: the Normal distribution for continuous and the log-normal distribution for positive-continuous outcomes. While employed Kropko (2013) employed the Normal distribution in a traditional IRT framework, no one has yet used the log-Normal distribution.

For a normal distribution, the ideal point model is used to predict the mean of Y_{ijt} given item-specific residual standard deviation σ_j :

$$L(Y_{ijt}|\alpha_{it}, \gamma_j, \beta_j, \sigma_j) = \prod_{i=1}^I \prod_{j=1}^J \prod_{t=1}^T \frac{1}{\sqrt{2\pi\sigma_j^2}} e^{-\frac{(Y_{ijt} - (\gamma_j\alpha_{it} - \beta_j))^2}{2\sigma_j^2}}$$

The log-normal distribution is the same distribution except that the domain of the distribution is on $\log(Y_{ijt})$:

$$L(Y_{ijt}|\alpha_{it}, \gamma_j, \beta_j, \sigma_j) = \prod_{i=1}^I \prod_{j=1}^J \prod_{t=1}^T \frac{1}{\sqrt{2\pi\sigma_j^2}} e^{-\frac{(\log(Y_{ijt}) - (\gamma_j\alpha_{it} - \beta_j))^2}{2\sigma_j^2}}$$

which naturally imposes the condition that $Y_{ijt} > 0$.

The Log-Normal parameterization is useful for continuous variables that are only defined on positive real numbers. For variables with substantial skew, such as income, this parameterization has a more useful interpretation than the more general Normal distribution. With both of these distributions, most continuous distributions can be modeled with reasonable accuracy.

The residual standard deviation σ_j is given a weakly informative exponential prior:

$$\sigma_j \sim E(1)$$

Practically speaking, unbounded continuous items should be standardized (mean 0 with an SD of 1) so that the range of the outcome remains reasonably close to zero. Because MCMC algorithms are geometric, variables with wide variation can diminish sampling efficiency. Furthermore, if other items are included with different scales, the nature of a shared linear model across items can increase model misfit substantially.

1.4 Ordered Beta Distribution

The ordered beta distribution is a robust model for dependent variables that are in the contained $[0, 1]$ interval (Kubinec 2022). Examples of these distributions in social science data include percentages, proportions, and sliders (such as those used in divide-the-pie experiments). The ordered beta distribution is implemented instead of the simpler beta distribution in order to allow for responses at the bounds; i.e., observations of 0 (or 0%) and 1 (or 100%). Such observations are quite common in social science data and the ordered beta distribution

only needs two additional parameters (ordered cutpoints) to model the full outcome. Without observations at the bounds of the scale, the ordered beta distribution effectively collapses to the beta distribution on the open interval $(0, 1)$.

The `idealstan` parameterization is as follows:

$$f(Y_{ijt} \mid \Omega, \Delta, \mu_j, \phi_j) = \begin{cases} \Omega & \text{if } Y_{ijt} = 0 \\ (1 - \Omega)\Delta & \text{if } Y_{ijt} = 1 \\ (1 - \Omega)(1 - \Delta)\text{Beta}(\mu_j, \phi_j) & \text{if } Y_{ijt} \in (0, 1) \end{cases}$$

The quantities Ω and Γ are the probabilities of the lower endpoint 0 and the probability of a response in the open interval $(0, 1)$. These probabilities are parameterized by ordered cutpoints $k \in \{1, 2\} : k_1 > k_2$. In other words, in the first stage of the model, either a continuous $(0, 1)$ or a discrete (0 or 1) outcome is selected, and then if the outcome is continuous, then it is also modeled with the Beta distribution.

The Beta distribution is further defined as follows:

$$f(Y_{ijt}, \omega_j, \tau_j) = \frac{\Gamma(\omega_j + \tau_j)}{\Gamma(\omega_j)\Gamma(\tau_j)} y_i^{\omega_j-1} (1 - Y_{ijt})^{\tau_j-1}$$

This Beta distribution is further reparameterized to allow for a linear model to predict the expected value (μ) while ϕ is the dispersion of the distribution. To do so, we must substitute for the shape parameters ω and τ :

$$\mu_j = \frac{\omega_j}{\omega_j + \tau_j}$$

$$\phi = \omega + \tau$$

Finally, to model $E[Y_{ijt}]$, we use the same IRT linear model as above:

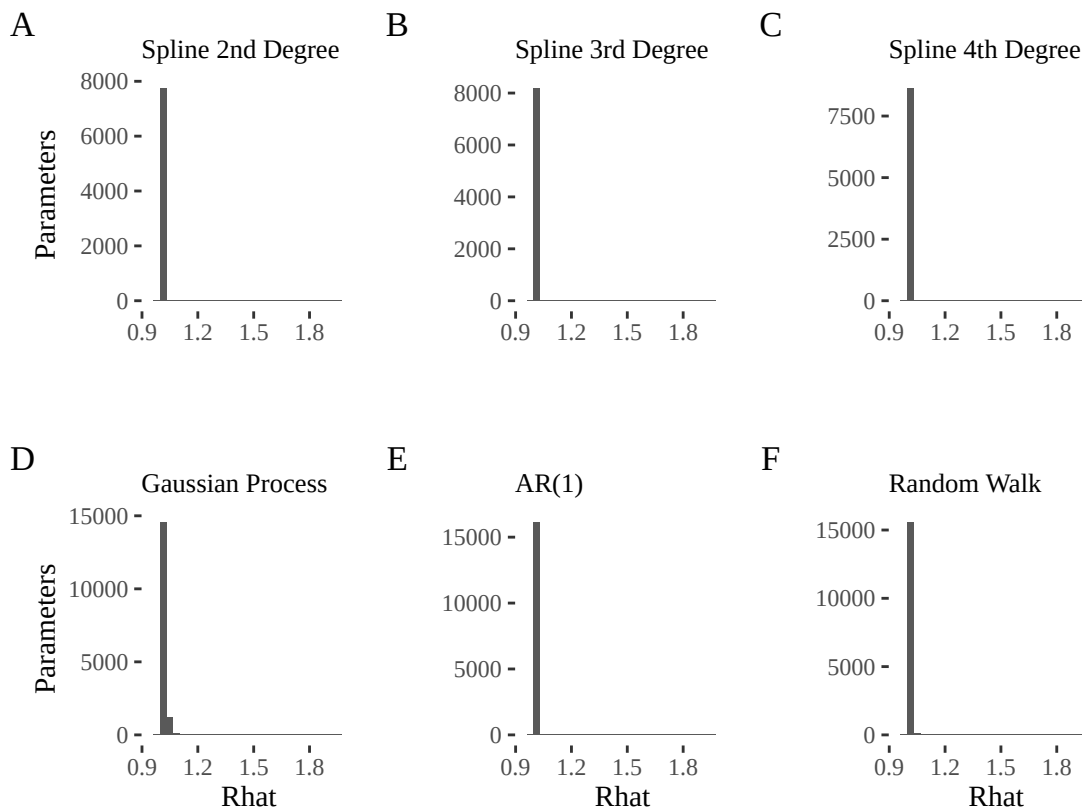
$$\mu_j = g(\gamma_j \alpha_{it} - \beta_j)$$

where $g(\cdot)$ is the inverse logit function.

For further definition of this model, see Kubinec (2022).

2 Model Convergence Diagnostics

In Figure Figure 1 I show the convergence diagnostics for all of the time-varying ideal point models of the 115th Congress. These convergence diagnostics employ the split-Rhat metric, which measures the ratio of variance of a Markov chain to itself over time and to other chains run independently. For this estimation, three chains were run with 500 post-warmup draws each, providing a difficult test for convergence. All of the models have almost all of their Rhat values below 1.1 with the exception of the Gaussian process model, which has a few parameters larger than 1.1. This more limited convergence indicates that the Gaussian process' much more demanding time-series function requires more draws than the other models to ensure a fully-explored posterior. However, even with 500 draws per chain, there is clear evidence of convergence for most parameters and inferences should be stable.



Plots show split-Rhat values from three independent Hamiltonian Markov Chain Monte Carlo chains with 500 post-warmup iterations per chain.

Figure 1: Comparison of Convergence Diagnostics Across Specifications

3 Parallelization Speed-up Simulation

In this section I report on simulations that examined the speed improvements with the use of multiple cores for within-chain parallelization of HMC gradient calculations. Figure 2 shows how estimation decreases for a given number of cores used for within-chain parallelization. The results for each core are the average across 2,000 Monte Carlo simulations of a binary/Bernoulli `idealstan` model with no missing data adjustment. For each draw, a number of persons between 10 and 200 and a number of items between 10 and 200 was selected so that the simulation includes substantial heterogeneity. Each time estimate in Figure 2 is an average across this heterogeneity and as such should be a reasonably robust estimate.

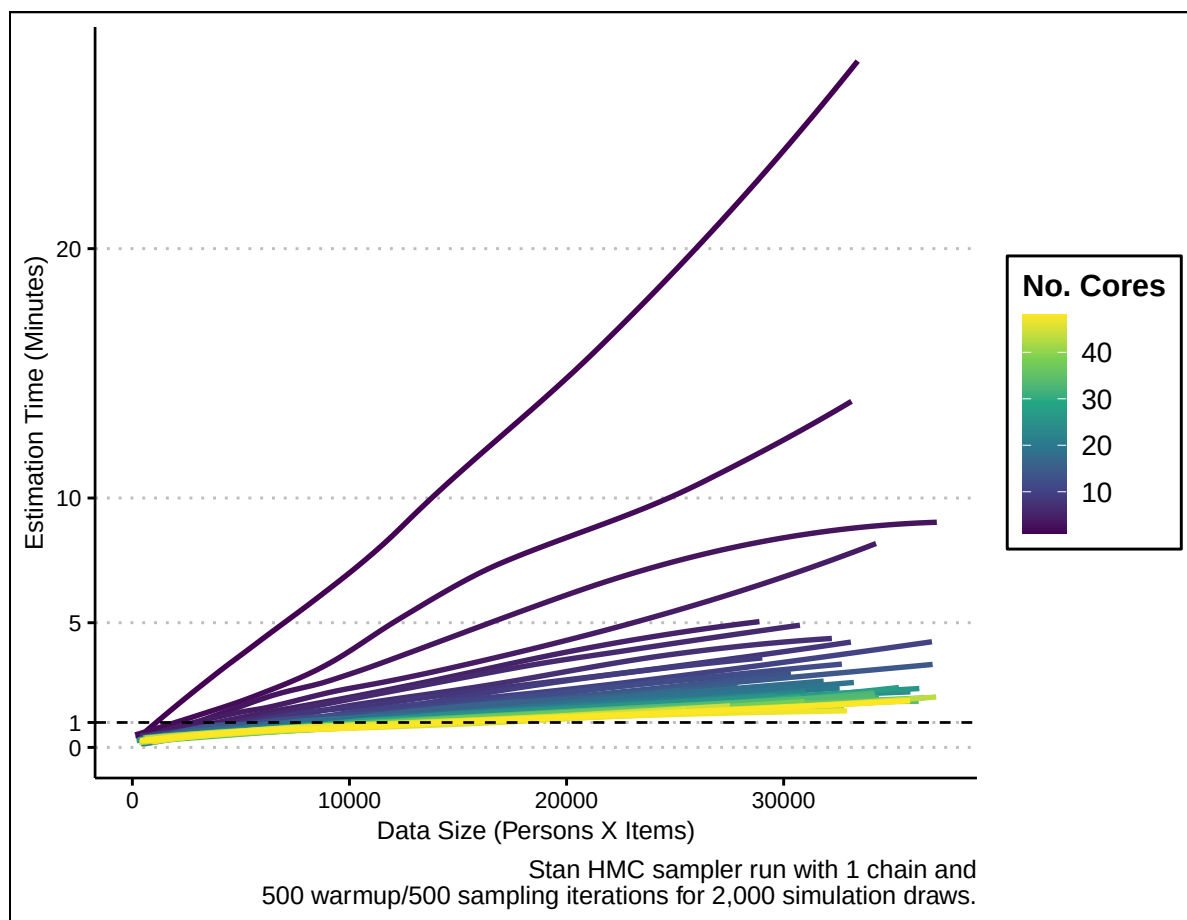


Figure 2: Estimation Time for Idealstan Models by No. of Cores

As can be seen in Figure 2, the most dramatic speedups are for moving from a single core to 2 to 4 cores. Above that number of cores, estimation time continues to decrease but at a decreasing rate (i.e. the second derivative is less than 0). This is a known behavior of parallel processing, especially when there is computational overhead from running multiple processes. For Markov

Chain Monte Carlo, it is not possible to run embarrassingly parallel processes in which there is a low fixed cost to using multiple cores. Rather, the process is partially parallelized via the computations required to calculate the gradients necessary for Hamiltonian Markov Chain Monte Carlo, i.e., where the algorithm should “jump” next.

The `idealstan` package attempts to minimize this overhead by using Stan’s map-reduce functionality to estimate the likelihood over subsets of data defined either by persons or items. For time-varying models, map-reduce must be performed over persons because the value of the person ideal point in time t depends on the prior value $t - 1$, and as a consequence, attempting to estimate the likelihood for each item separately would yield biased results. However, using map-reduce over time-varying trajectories is often optimal in any case as the time-varying parameters for each person are calculated separately in each process, permitting less computational overhead.

For static models, the user can choose to parallelize over either items or persons. There is no clear guideline for which may be faster, and in general the most important criterion may be the number of items or persons relative to cores. For example, if there are 5 persons and 100 items in the data but only 5 cores, it would make the most sense to use persons as that would yield an efficient parallelization scheme.

4 Absences in the U.S. Congress

As described in the main text, Figure 3 and Figure 4 show rates of absences over time and within each session in the U.S. House.

5 Additional Simulation Results

We show the average coverage of true ideal points by different models in Figure 5 and the average number of sign rotations in Figure 6. By sign rotation, we mean the number of times when the RMSE of the inverse estimates was lower than the RMSE of the estimates, in which case we would multiply the estimates by -1 to account for the model converging to the wrong mode.

Figure 5 shows that no one model has perfect coverage, as might be expected given that coverage is calculated from standardized ideal points, which can induce bias. However, `idealstan` with HMC comes quite close except for the Gaussian process model. Generally speaking, the Laplace approximation and `emIRT` have the best coverage for non-Monte Carlo methods, though coverage is better for simpler models and when there is no missing data.

The average number of sign rotations in Figure 6 reveals that more complex models have higher number of sign rotations on average. Gaussian process models with `idealstan` have higher sign rotation rates than splines or random walk models, while models that incorporate missing

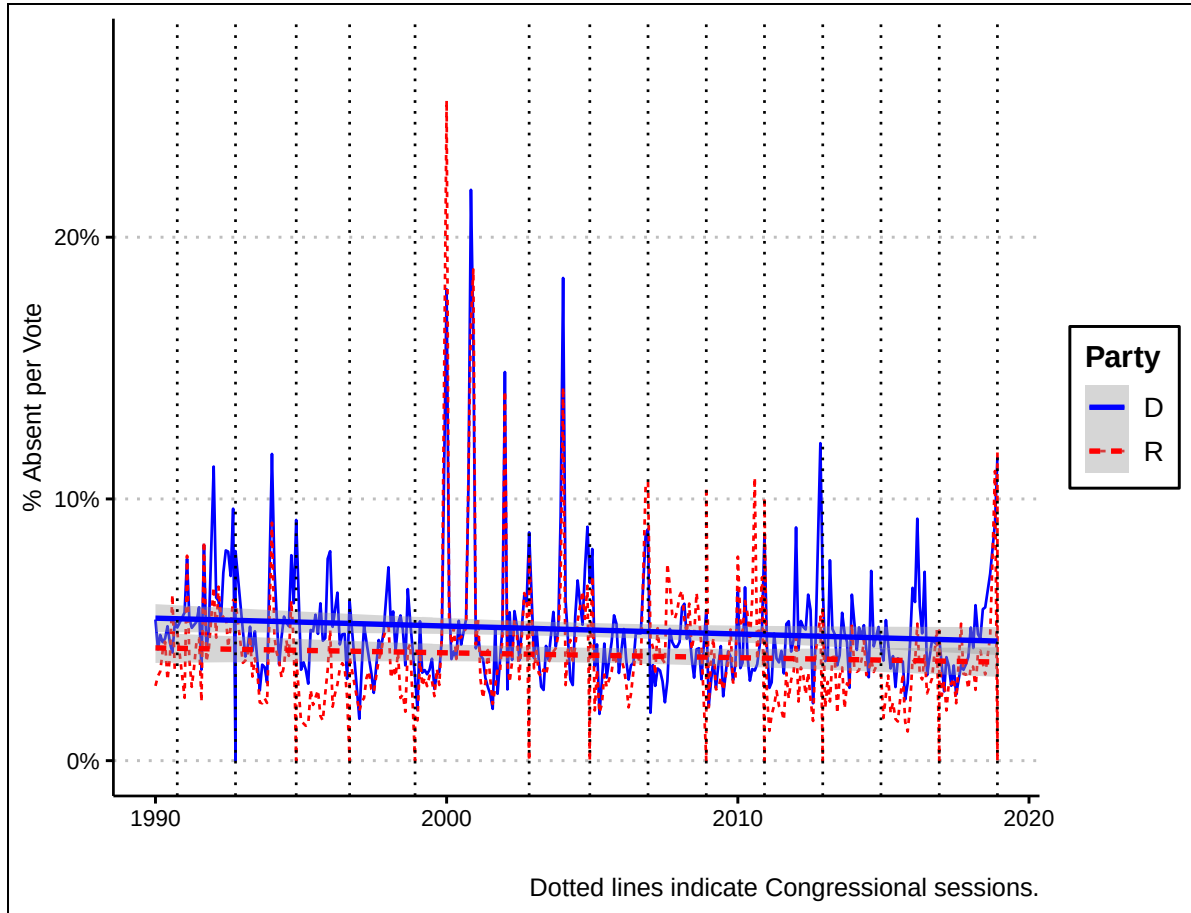


Figure 3: Time-varying Absence Rates in the U.S. House

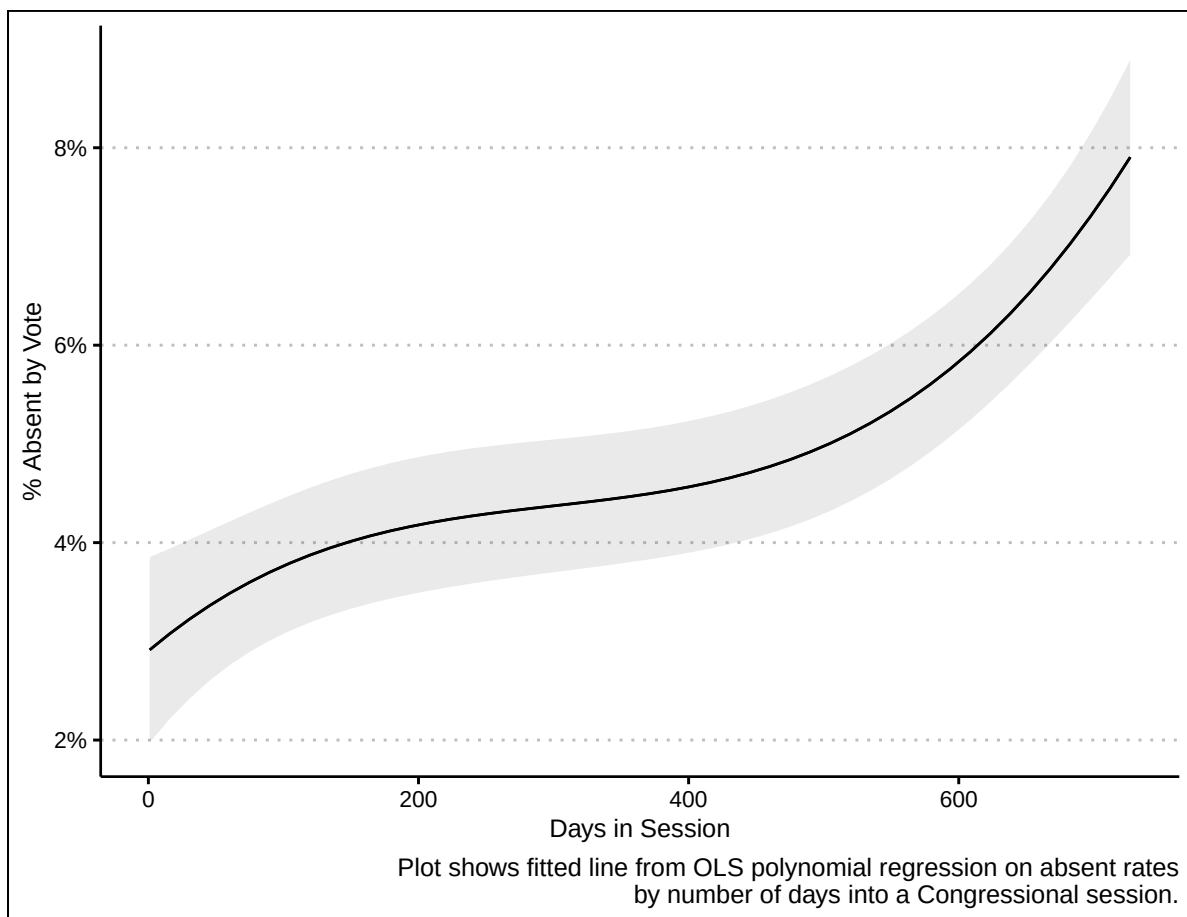


Figure 4: Ordinary Least Squares Polynomial Regression of Days in Session on Absence Rates in the U.S. House

data also have higher sign rotation rates. While it is important to keep these metrics in mind, these models are designed to be employed for specific measurement problems where scholars have ample expertise to identify multiple items or persons to constrain for identification. At the same time, as discussed in the empirical section of the main paper, there are cases where a simpler model like splines can yield fewer sign rotations even without additional identifying information for the model.

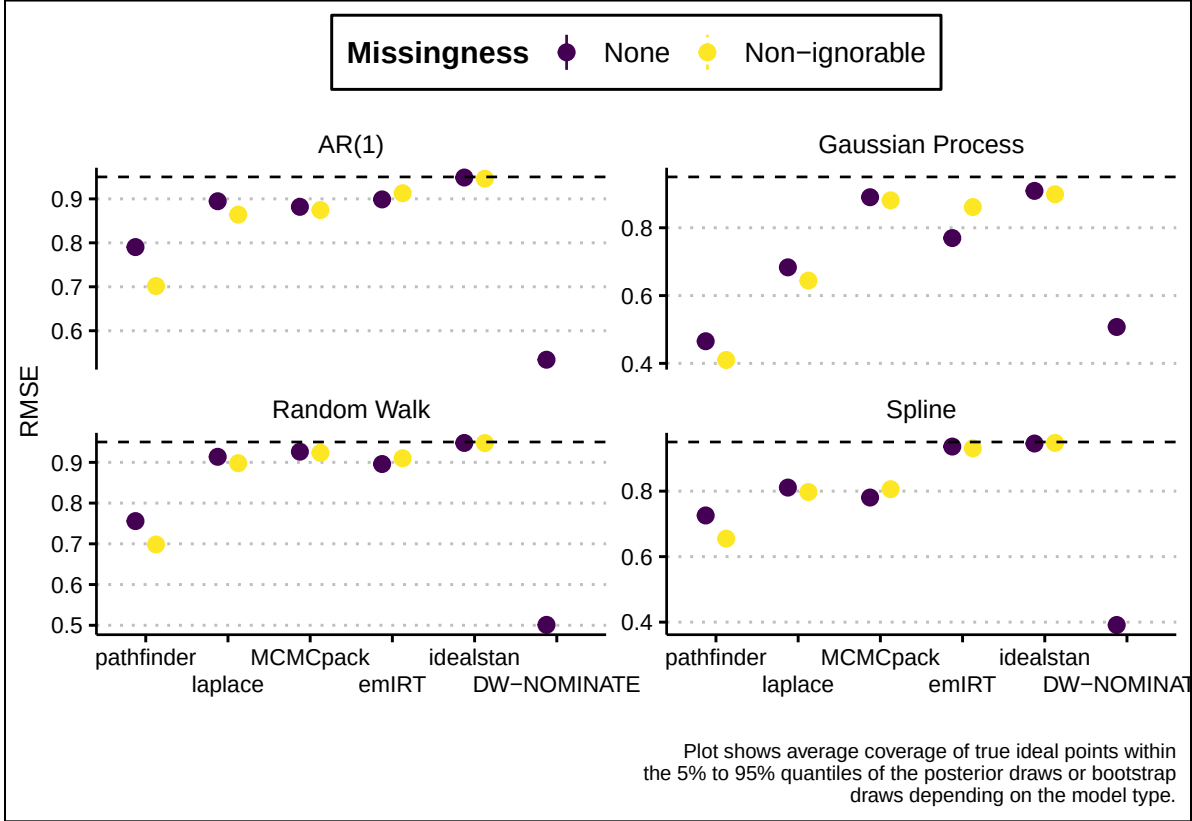


Figure 5: Average 5% to 95% Coverage of True Ideal Points by Model and Time Series Process

6 Priors, Identification, and Starting Values

6.1 Model Priors

Identification for models with time-varying α_{it} is a challenging endeavor that lacks a comprehensive discussion in the literature, often leaving practitioners to spend considerable time guessing what settings might produce an identified latent variable. Martin and Quinn (2002) addressed this issue for random-walk models by restricting the variance σ_i to values strictly

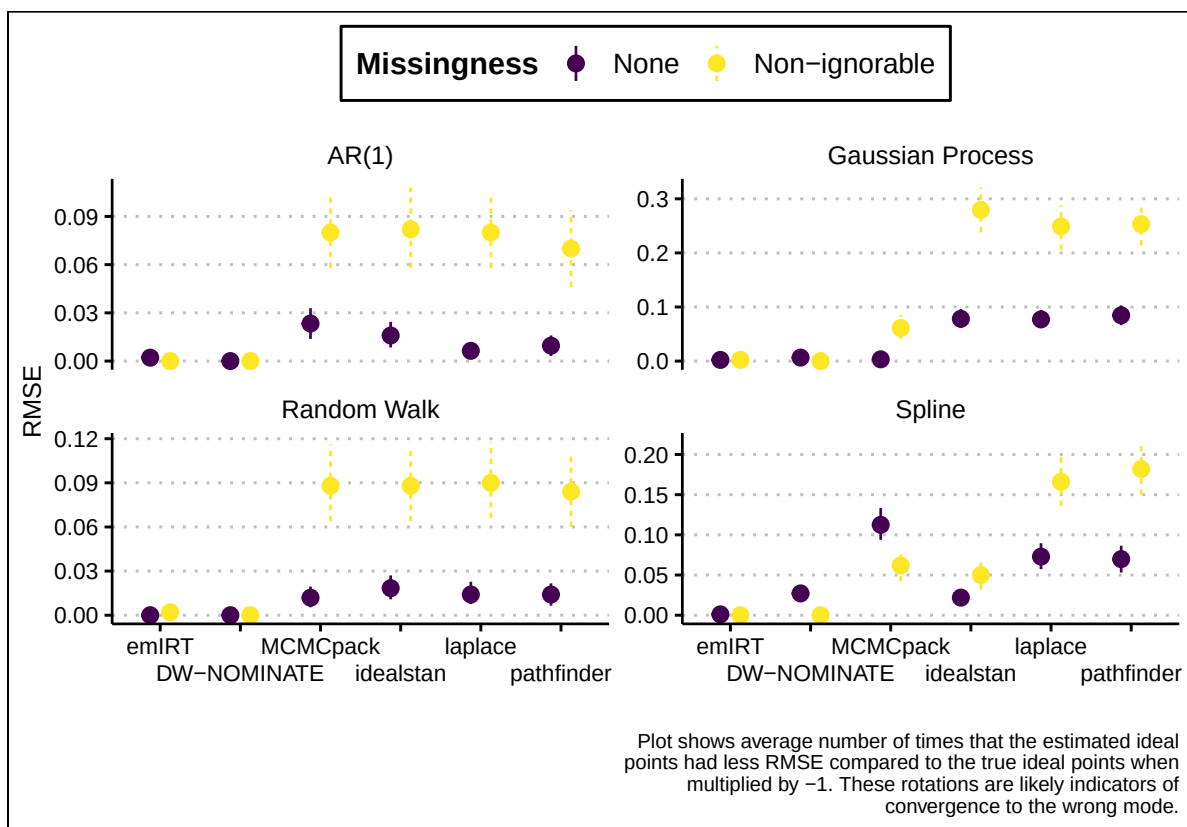


Figure 6: Average Proportion of Sign Rotations By Model and Time Series Process

between 0 and 0.1. This hard-coded constraint forced the time series to change little over time, though it also limits inference as the time series can only differ so much from each other.

By contrast, I impose generally diffuse priors, or priors strong enough to impose some scaling identifiability, without forcing hard constraints on variance or other parameters. Rather than employ priors that try to fix the ideal point estimation, I argue that an appropriately theoretical model structure should allow for relatively weak priors, which should only be constrained to include relevant empirical information like the known ideological position of actors or items. When the data are both sparse and noisy, and time-varying estimation becomes non-trivial, then the model should employ a time series function like a 3rd-degree spline to remove the possibility of theoretically impossible shifts in ideal points over time. I discuss this approach in more detail in the empirical application and in the simulation results.

I define the priors in terms of the distributions assigned to each parameter in θ :

$$\alpha_i \sim \text{Normal}(0, 3) \tag{4}$$

$$\gamma_j \sim \text{GeneralizedBeta}(2, 2) \tag{5}$$

$$\beta_j \sim \text{Normal}(0, 3) \tag{6}$$

The ideal points α_i and the discrimination parameters β_j are given a loose $N(0, 3)$ prior that is weakly informative for most scales and is only needed to make the posterior proper. Identification of the posterior, which is an important issue as these types of measurement models can have multiple possible solutions with equal likelihood (Bafumi et al. 2005), happens via a Generalized Beta prior on the discrimination parameters γ_j :

$$f(x; \alpha, \beta, a, b) = \frac{(x - a)^{\alpha-1} (b - x)^{\beta-1}}{(b - a)^{\alpha+\beta-1} B(\alpha, \beta)} \tag{7}$$

where:

- $\alpha > 0$ and $\beta > 0$ are the shape and scale parameters, analogous to the prior sample size,
- $a = -1$ and $b = 1$ define the support of the distribution,
- $B(\alpha, \beta) = \int_0^1 t^{\alpha-1} (1 - t)^{\beta-1} dt$ is the Beta function.

The reason I use the Generalized Beta distribution for the items is to resolve two problems at once. First, the restrictions on the range of the γ_j parameters solve the problem of identifying a scale for the latent variable because the item discrimination parameters must stay within this range. The Generalized Beta distribution further allows this range to be both positive and negative, which is required for the ideal point model to estimate latent variables with

bi-polarity (such as both conservative and liberal ideology). To identify the rotation of the ideal points, specific items can be given a very tight prior, such as:

$$\gamma_{j=k} \sim \text{GeneralizedBeta}(10, 1000)$$

for a parameter with the majority its support around the point -0.98 and

$$\gamma_{j=k'} \sim \text{GeneralizedBeta}(1000, 10)$$

for a parameter with the majority of its support around the point +0.98. The shape and scale parameters of the Generalized Beta have an intuitive interpretation as they represent the prior observed negative and positive observations for a given proportion. As a result, increasing both the scale and shape proportionally will preserve the sample expected value but reduce uncertainty around the expected value of the distribution. With larger datasets, increasing both the shape and scale to larger values, such as (20000, 200), is necessary as the strength of the prior is relative to the size of the data as in main text Equation 1. To fit a traditional IRT model, on the other hand, all this is necessary is to modify the lower bound (a) of the Generalized Beta distribution to 0 rather than -1.

Constraining the discrimination parameters builds on work that shows that identification of the items (i.e., votes) in an ideal point model has superior theoretical properties as it allows for the ideal points to float freely (Morucci et al. 2024). In addition, constraining item parameters is helpful for dynamic ideal point estimation as items may occur at different points in the time series, whereas constraining dynamic ideal points is limited to fixing the starting positions of time series. With increasing time points, the starting positions of the actors become less influential, making identification more difficult to achieve.

I assign weakly informative values of +2 to each of the distribution’s parameters (scale and shape) so that values of 0, or the midpoint of the distribution, are weakly preferred over the end points. With this prior, no further work is necessary to fix a scale and all other parameters can have weakly informative priors. In fact, if there is enough data, the discrimination parameters can have the uninformative Generalized Beta prior of +1, +1 so that all unconstrained discrimination parameters γ_j are equally likely in the $(-1, 1)$ interval—which is much less informative than the Normal prior that is commonly used.

If the missing data is modeled directly as described in the main text, the model does not generally require more identifying information than previously stipulated (i.e., at least some observed item discriminations γ_j pinned to high and low values). However, the missingness space does raise model complexity, it can require more computation and can in some cases require additional pinned item discrimination parameters. However, the missingness parameters themselves can have similar weakly informative Generalized Beta priors without causing any serious problems with identification:

$$\nu_j \sim \text{GeneralizedBeta}(2, 2) \quad (8)$$

$$\omega_j \sim \text{Normal}(0, 3) \quad (9)$$

6.2 Starting Values

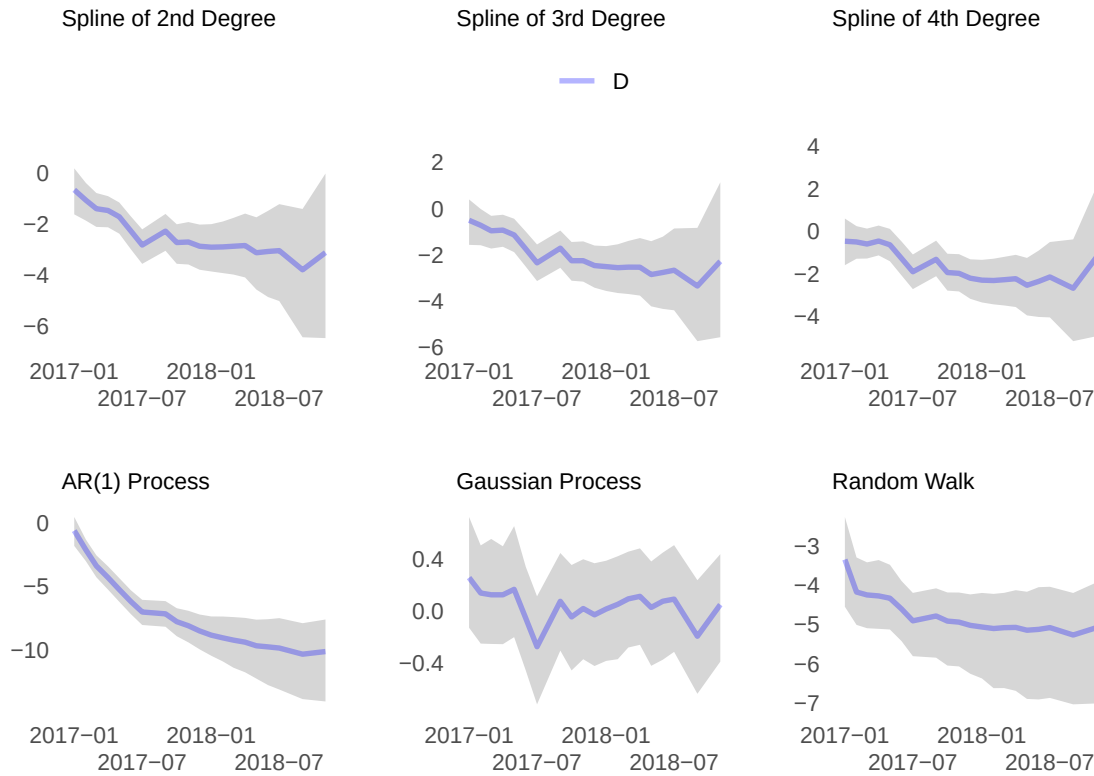
Identifying a single mode (or estimate) of time-varying latent variables given noisy data can be very difficult to achieve. As mentioned previously, Martin and Quinn (2002) used a very restrictive bound on over-time variance to identify their estimates of Supreme Court justices. In order to estimate DW-NOMINATE using optimization methods, Poole and Rosenthal had to undergo a laborious process of choosing starting values for legislator ideal points, estimating dynamic scores, and then reviewing the scores to see if they resembled their theoretical priors (Carroll et al. 2009). Similarly, Barberá (2015) scores were only possible to derive by using starting values for HMC with Stan that accorded with theoretical expectations about ideal point distributions.

While identifying the sign and rotation of ideal points with pinned discrimination parameters is sufficient for static models, high-dimensional time-varying latent variable model posteriors can be difficult for HMC to adequately explore even though it is the most robust and stable MCMC algorithm available (Carpenter et al. 2017). For high-dimensional models, such as latent variables with hundreds of time points, HMC can end up in lower-probability modes that may have the opposite polarity of what the identification restrictions require. Specifying starting values that are closer to the posterior mode can help HMC converge and explore the posterior for very difficult problems (Barberá 2015). However, specifying starting values a priori can either be extremely tedious as the DW-NOMINATE procedure mentioned previously or it can result in latent variables that depend on arbitrarily chosen starting values that can mask convergence issues with HMC and result in fragile inferences.

To address this problem, **idealstan** implements a novel method for identifying starting values for high-dimensional HMC using what is known as the Pathfinder algorithm (Zhang et al. 2021). The Pathfinder algorithm is a form of variational inference (Goplerud 2019; Grimmer 2011) that approximates the joint posterior by minimizing the Kullback-Leibler divergence with a form of Newton optimization. While this approximation is unlikely to capture the exactly correct form of the joint posterior distribution, it is remarkably robust considering its relative simplicity (Zhang et al. 2021). Importantly, Pathfinder integrates with the Stan framework so that Pathfinder can first be used to identify starting values for the HMC algorithm, which is particularly important for high-dimensional time-varying models. These starting values are optimal in the sense that they should be reasonably close to the posterior mode, and they also do not reflect researcher discretion, which makes them a more robust choice for measurement exercises as they are in principle replicable.

7 Additional Figures

Please see below for additional figures related to the estimation of ideal points in the U.S. House as presented in the main text.



Posterior average values shown for 115th Congress legislator ideal points.
Item discrimination for Republican party-line votes was constrained to be positive.

Figure 7: Comparison of Tim Walz Time-varying Ideal Points for the 115th Congress with Observed Votes

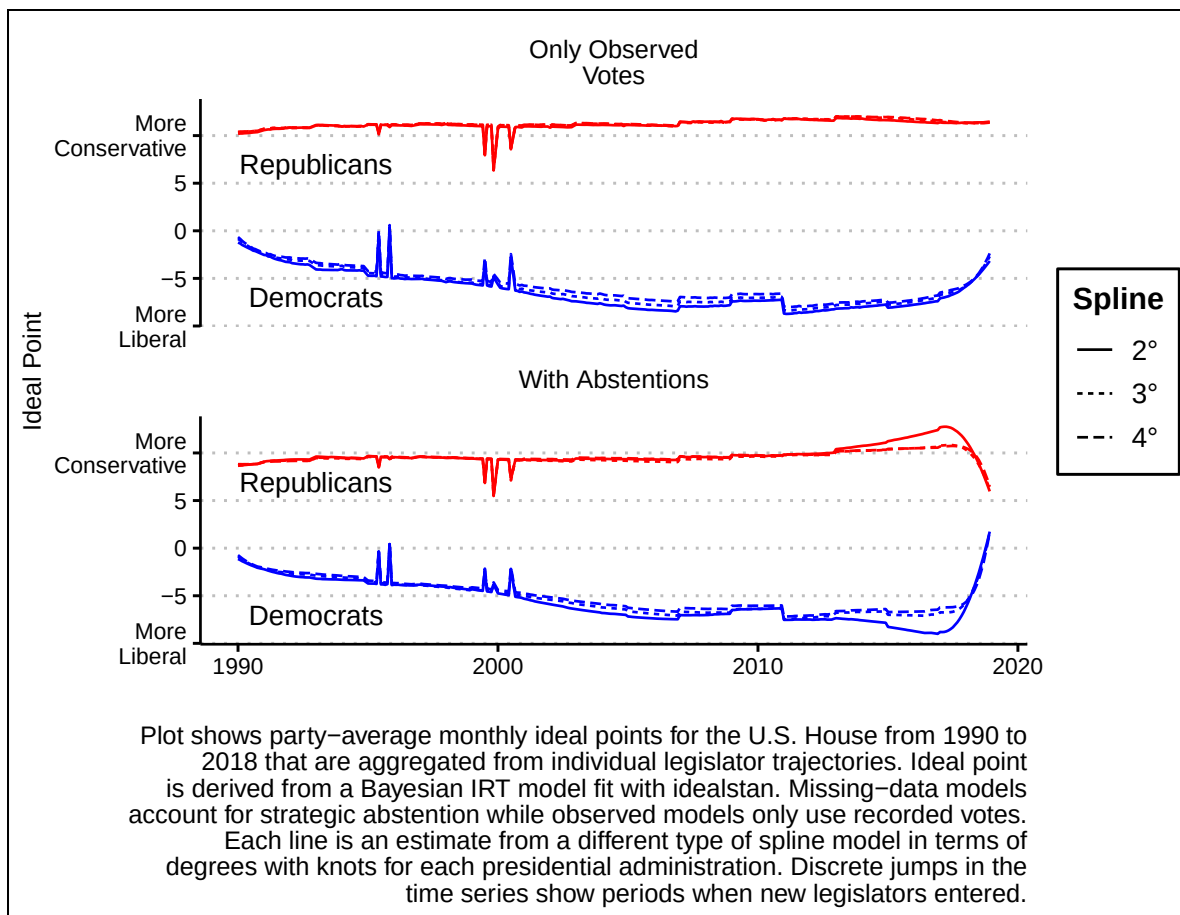
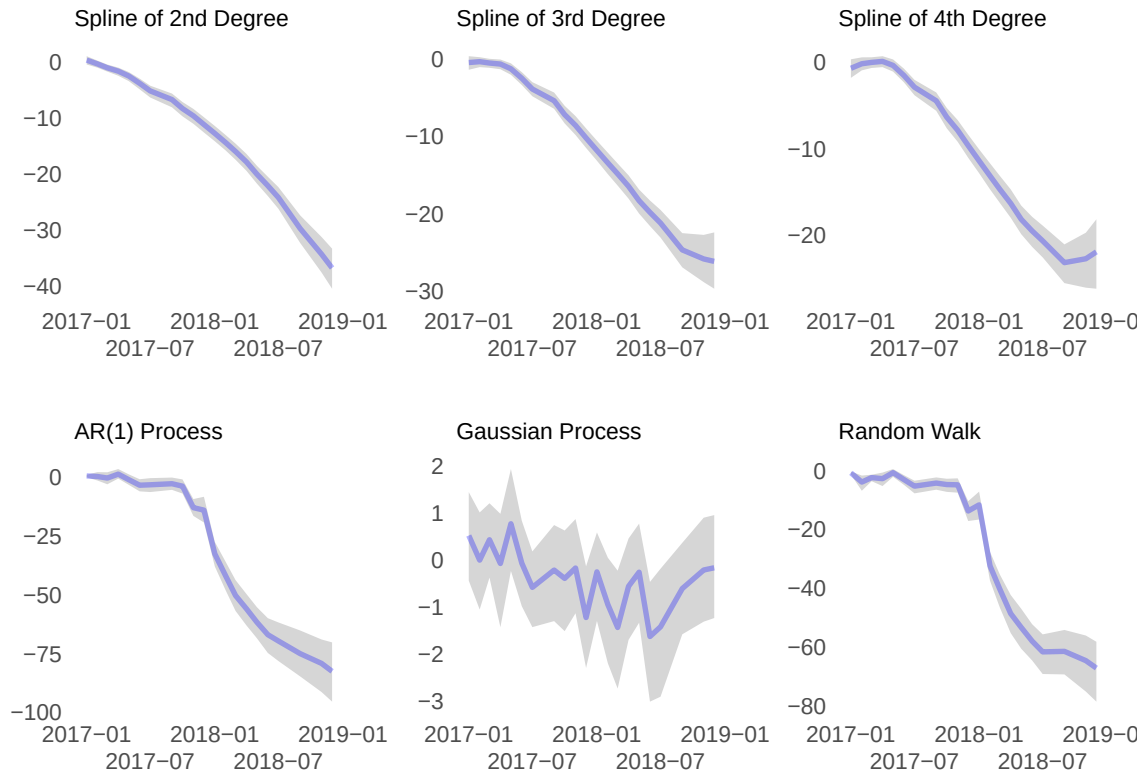


Figure 8: Monthly Party-level Ideal Points for the U.S. House



Posterior average values shown for 115th Congress legislator ideal points.
Item discrimination for Republican party-line votes was constrained to be positive.

Figure 9: Comparison of Tim Walz Time-varying Ideal Points for the 115th Congress with Missing Votes

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